History-Conditioned SAC under Sensor Faults: A Pilot Study of Adaptation and Retention

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Why This Research? The Origin & Motivation

Training Plateaus in Real-Time TrackMania 2020 and the Mystery of Sensory Shifts

- **The High-Speed Challenge:** Our overarching goal is real-time visual control in TrackMania 2020 at 100–500 Hz. In this domain, sequence models and world models repeatedly encountered severe performance plateaus.
- **The Diagnostic Dilemma:** Why did policies stall? Did they suffer from loss of plasticity (inability to learn further)? Primacy bias (overfitting early data)? Or catastrophic forgetting during fine-tuning?
- **The Physical Reality of Sensors:** Physical deployment introduces unannounced sensory degradation: camera vibration (noise) and transmission lag (delay). Actuator faults change control; sensor faults change information.
- **Need for Controlled Isolation:** In end-to-end visual driving, these factors are deeply entangled. We needed a rigorous, controlled MuJoCo benchmark to isolate and measure adaptation and retention directly.

(a) Motivating application: TrackMania 2020



Real-time visual control

Goal: learn new conditions while retaining earlier control behavior.

Saved development capture

(b) Executed study: HalfCheetah-v5 and Ant-v5

16 source policies per task
2 controllers x 8 source seeds

Old condition A: clean
2 replay episodes per source

Five branches per source
Noise or 3-step delay x $\rho \in \{0, 0.5\}$
plus a clean-training control

Continued SAC updates
20,000 new steps, 5,000 updates
ρ = old-replay batch fraction

Freeze the updated checkpoint
At 0, 2k, 4k, 8k, 12k, 16k and 20k new steps

Evaluate A: retention
Clean return; critic residual

Evaluate B: adaptation
Sensor return; critic residual

Same checkpoint for A and B; evaluation episodes stay out of replay.

Core Research Questions (What We Set Out to Answer)

Decoupling Adaptation, Catastrophic Forgetting, and Replay Buffer Dynamics

- **RQ1: Adaptation vs. Retention Trade-off:** When a trained policy encounters unannounced sensory degradation, can continued training restore return (Adaptation), and does fine-tuning erase performance on nominal clean conditions (Retention)?
- **RQ2: Architectural Inductive Bias:** Does history conditioning via a sequence model (Causal Transformer) provide an architectural buffer against sensor faults compared to a memoryless reactive MLP, or does past context misinform under latency?
- **RQ3: The Replay Buffer Dilemma:** Does the standard continual learning remedy—mixing prior clean experience into replay—prevent forgetting, or does it introduce an off-policy distribution mismatch that corrupts Bellman value targets?
- **RQ4: Real-Time Edge Feasibility:** Can an expressive history-conditioned transformer execute within the strict sub-2ms cycle budget required for 500 Hz physical robotic control on edge CPU hardware?

Experimental Formulation & Fault Modeling

Partially Observable Formulation (POMDP) and Controlled 5-Branch Protocol

- **POMDP Formulation:** Continuous control modeled as M = (S, A, T, R, Omega, O, gamma), where the policy receives degraded observations s_tilde_t rather than ground-truth physical states.
- **Fault Model 1: Gaussian Noise:** s_tilde_t = s_t + e_t, e_t ~ N(0, sigma^2 I) with sigma = 0.15. Perturbs signal amplitude without introducing temporal lag.
- **Fault Model 2: Observation Delay:** FIFO ring buffer of depth d = 3 (s_tilde_t = s_t-d). Breaks Markovian velocity tracking; observation represents outdated physical plant state.
- **5-Branch Experimental Design:** 1. Clean Control (isolates optimization drift)

2. Noise Only (pure adaptation)

3. Noise + Replay (rehearsal)

4. Delay Only (pure adaptation)

5. Delay + Replay (rehearsal)

- **Paired-Seed Statistical Rigour:** 8 independent source seeds per controller and task (160 policy branches, 20,000 steps each). Exact Wilcoxon signed-rank tests with Holm-Bonferroni correction.

Architectural Inductive Biases: Memory vs. Reactivity

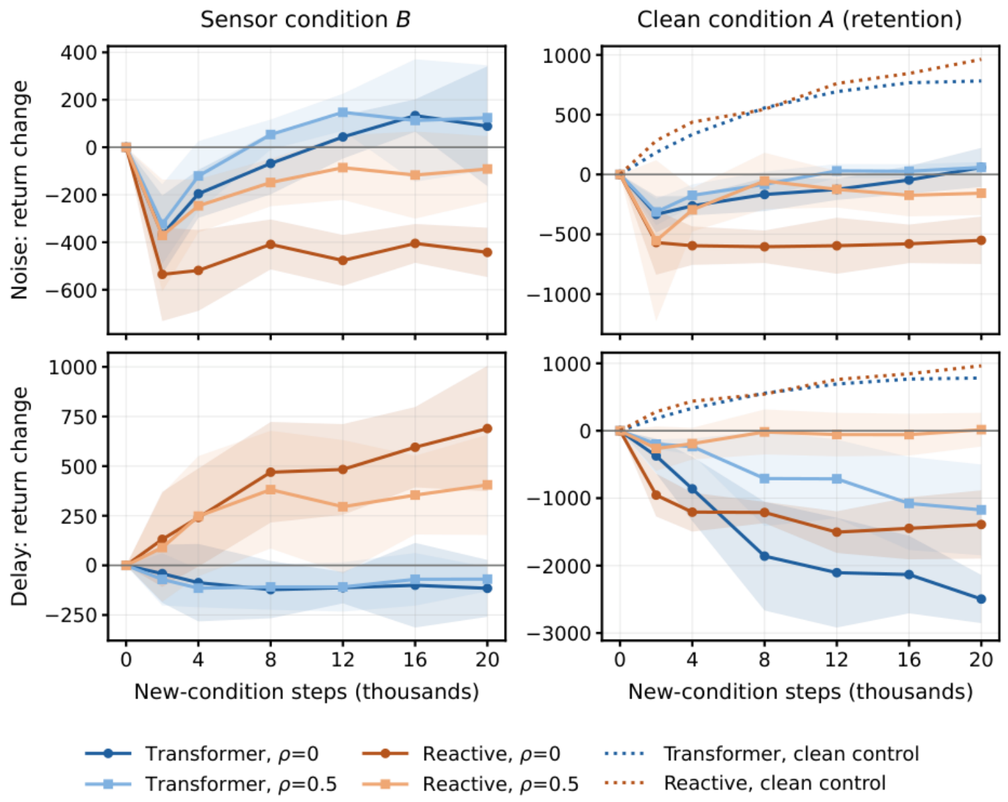
History-Conditioned Causal Transformer vs. Reactive MLP Baseline

- **Reactive Controller Baseline (MLP):** Standard SAC policy: 2 hidden layers with 256 units each, ReLU activations. 71.9k actor / 72.5k critic parameters. Memoryless mapping s_t -> a_t, strictly reliant on instantaneous input.
- **History-Conditioned Controller (Transformer):** GPT-style causal transformer: sliding context window K = 10 steps, token embedding d_model = 64, 2 causal self-attention blocks (4 heads), LayerNorm, and MLP projection head (87.6k actor / 92.3k critic parameters).
- **In-Context Boundary Handling:** Context resets at episode boundaries. The model observes historical trajectory segments; cross-episode adaptation occurs exclusively through continued parameter updates.
- **Replay Mixing Rehearsal:** 50% nominal clean transitions mixed with 50% online corrupted transitions to investigate whether memory preservation impairs adaptation velocity.

Empirical Findings: HalfCheetah-v5 Dynamics

High-Frequency Noise Filtering vs. Catastrophic Observation Delay Collapse

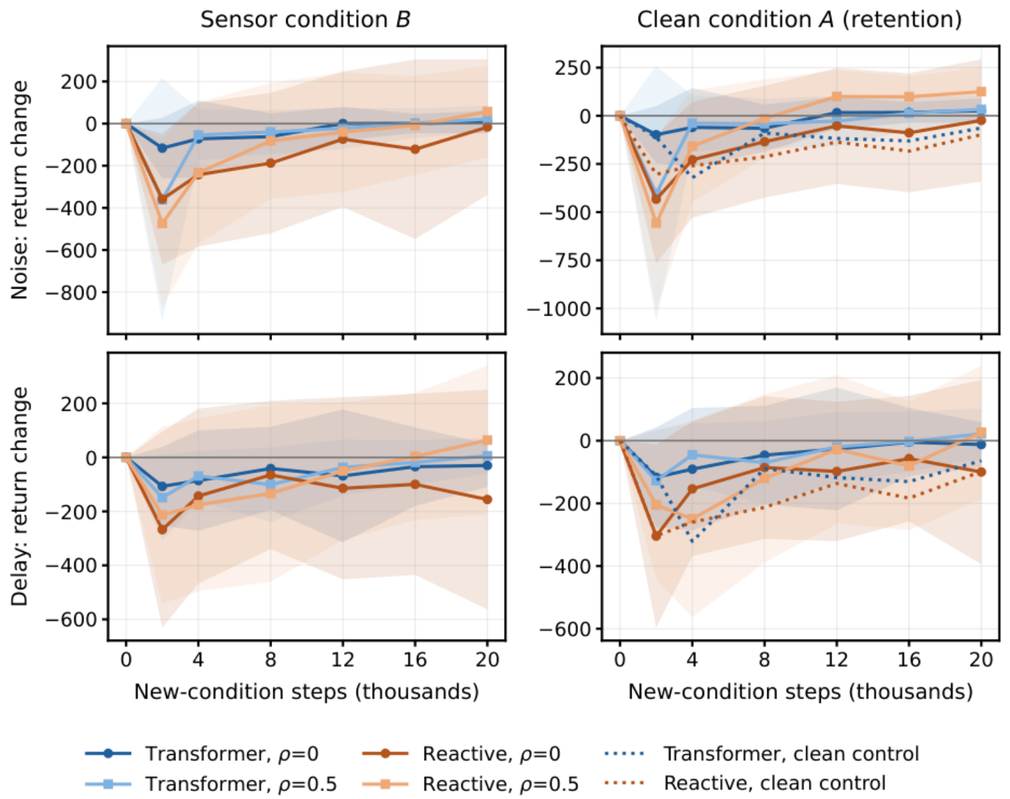
- **Observation 1: Noise Resilience:** Under Gaussian sensor noise, both reactive MLP and transformer adapt swiftly, recovering near-optimal velocity. Replay mixing successfully prevents catastrophic forgetting with minimal penalty.
- **Observation 2: The Severe Delay Crash:** Observation delay causes immediate, severe performance collapse (>2,000 return penalty). Continued training achieves only partial recovery across 20,000 steps.
- **The Transformer Delay Paradox:** Under delay, the transformer actually struggled more than the reactive MLP! Causal self-attention attended to stale past tokens that misrepresented current plant physics, amplifying control latency.



Empirical Findings: Ant-v5 & Morphological Sensitivity

Morphological Standing Posture and the Warm-Start Re-Optimization Dip

- **Observation 3: Morphological Immunity:** Under the standardized 400,000-step source recipe, all 8 Ant source seeds converged to an ultra-stable standing posture (mean forward velocity +/-0.007 m/s).
- **Negligible Fault Sensitivity:** Because standing requires minimal dynamic velocity feedback, sensor noise and delay caused virtually zero return penalty. Reported transparently rather than retroactively tuning hyperparameters.
- **Observation 4: Re-Optimization Transient:** All Ant branches (including clean controls) dipped in return during the first 2,000 steps before recovering. Scientific Insight: This transient dip is caused by re-starting the Adam optimizer, NOT by sensor faults!



Mechanistic Critic Diagnostic: Soft Bellman Target Distortion

Explaining Why Replay Buffer Mixing Hurt Adaptation Under Observation Delay

- **Diagnostic Framework:** Both soft Q-critics evaluated on held-out clean and corrupted trajectories at source checkpoint and after 20,000 steps. Evaluated soft Bellman residual and Monte-Carlo value-bias gap (Bias_corrupt - Bias_clean).
- **Massive Source Value Overestimation:** Source critics severely overvalued delayed history states (+171 and +268 return units in HalfCheetah).
- **Distortion From Replay Mixing:** Training on corrupted data alone shrank the bias gap to +27 and +114. Mixing 50% clean replay maintained the gap elevated at +89 and +204 (paired effect: +62 [16, 109] and +90 [41, 139], 7/8 seeds positive).
- **Theoretical Explanation:** Mixing stale clean transitions creates an off-policy distribution mismatch that actively distorts the soft Bellman target, proving that naive rehearsal harms adaptation under non-Markovian delay.

Real-Time Edge Viability: 500 Hz Benchmark

Sub-Millisecond Inference Proves Feasibility for High-Speed Robotic Control

Platform	Controller	Threads	Mean Latency	99th %ile	Max Latency	500 Hz Viable?
CPU (i7-8700)	Transformer Actor	1 thread	0.58 ms	0.66 ms	1.56 ms	YES (Budget < 2.0 ms)
CPU (i7-8700)	Reactive MLP Actor	1 thread	0.09 ms	0.12 ms	0.28 ms	YES
CPU (i7-8700)	Transformer Actor	4 threads	0.64 ms	0.78 ms	1.82 ms	YES
GPU (RTX 3070 Ti)	Transformer Actor	CUDA	1.64 ms	1.92 ms	3.14 ms	MARGINAL (Kernel overhead)

- **10,000 Timed Forward Passes:** Measured at deployment batch size 1 with real model inputs (excluding physics engine simulation overhead).
- **Strict Real-Time Compliance:** Transformer inference on single-threaded CPU consistently completes in under 1.56 ms, comfortably within the 2.0 ms cycle budget of 500 Hz robotics.
- **Edge Architecture Takeaway:** Single-threaded CPU execution outperforms GPU at batch size 1, confirming suitability for embedded robot microcontrollers without dedicated discrete accelerators.

Answers to the Research Questions: Key Takeaways

Actionable Scientific Insights for Fault-Tolerant Reinforcement Learning

- **Answer to RQ1 (Adaptation vs Retention):** Adaptation is fast and complete under high-frequency noise with full retention, but severely degraded under delay where nominal competence is permanently impaired.
- **Answer to RQ2 (Transformer Inductive Bias):** History conditioning acts as a natural temporal low-pass filter under noise, but exacerbates vulnerability under delay unless delay is explicitly compensated in the state representation.
- **Answer to RQ3 (Replay Buffer Trade-off):** Replay buffer mixing is not a universal panacea: it preserves nominal skills under noise, but induces severe soft Bellman target distortion under non-Markovian delay.
- **Answer to RQ4 (High-Rate Control Viability):** History-conditioned transformers require only 87k parameters and execute in 0.58 ms on a single CPU thread, satisfying real-time 500 Hz robotics.

The Path Forward & Open Science

Returning to TrackMania 2020, Physical Robotics, and Open Artifact Release

- **Implications for TrackMania & Robotics:** To overcome real-world delay, future controllers must incorporate explicit delay-aware predictive heads, temporal attention discounting, and dynamic uncertainty-weighted replay sampling.
- **Scalability Outlook:** Next phase investigates scaling these insights to high-dimensional continuous control (Humanoid-v5) and vision-based end-to-end TrackMania pipelines.
- **Full Open Science Release:** Complete reproducible codebase, sensor fault wrappers, frozen seed protocols, evaluation records, and checkpoints available at: https://github.com/Fane-Nathan/tmrl
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